

CYBOX-S: OPERATOR GUIDE & PRACTICAL WORKFLOW REFERENCE

High-Speed Offline Penetration Testing Worksheet · eJPTv2 / eCPPT Practical Companion

What is CYBOX-S?

CYBOX-S is a fast, keyboard-driven terminal worksheet and operational cockpit designed to eliminate exam cognitive overload. It provides an offline state machine for host discovery, port tracking, credential reuse, syntax cheatsheets, and question proofs. **It does not run autonomous exploits or rely on AI.** You retain 100% human control while CYBOX-S manages your operational memory.

INE eJPT / eCPPT Exam Compliance

- **100% Passive & Offline:** Runs entirely on localhost via local SQLite database.
- **Zero Autonomous Action:** Commands are copied to your clipboard; YOU execute them.
- **Zero Cloud / AI Dependencies:** No external API calls, leaks, or prohibited LLMs.
- **Permitted Personal Notes:** Functions strictly as a local worksheet and syntax lookup.

1. The Four Operational Stations (Switch with Keys 1, 2, 3, 4)

Station	Name & Purpose	What You See & Do Here	Hotkey
Station 1	Cockpit (Workbench) Primary attack surface station	Targets tree, open ports, service status triage, methodology checklist, field notes, and interactive command console.	[1]
Station 2	Playbook Browser Methodology & cheatsheet search	Browse full tactical playbooks (eJPT workflow, Web App OWASP, Active Directory, Pivoting, Linux & Windows PrivEsc). Search commands with instant copy.	[2]
Station 3	Credential Matrix 2D Vault & Lateral Spray Grid	2D grid of discovered credentials (rows) × target services (cols). Press [Enter] to compile & copy spray commands; press [Space] to cycle verification state.	[3]
Station 4	Loot & Flags Ledger Proof hashes & failure logs	View user/root flags, hashes, credentials found, and Failure Logs / Rabbit Holes . Ensures you never forget where you got stuck or what you discovered.	[4]

2. Station 1 Anatomy & Screen Real Estate

Cockpit is mathematically arranged in a zero-overflow 4-quadrant layout with a dedicated sidebar and bottom console bar:

SIDEBAR (Left 28% width)

- **ATTACK SURFACE:** Collapsible tree of target IPs and hostnames. Displays live service badges.
 - **CREDENTIALS PREVIEW:** Discovered usernames and masked passwords. Press **[Space]** to reveal.
- Tip: Press [b] anytime to toggle/collapse sidebar and give 100% width to Workbench!*

WORKBENCH (Right 72% width)

- **SERVICES & PORTS (Top):** Port, protocol, version, and potential. Press **[Space]** to cycle triage status. Highlight service to auto-filter checklist!
- **METHODOLOGY (Lower-Left):** Step-by-step checklist with progress bar. Auto-tracks current port.
- **NOTES & FINDINGS (Lower-Right):** Live scratchpad for leads, vulns, and flags.

BOTTOM CONSOLE BAR (Full Width) — Fast-capture command line & live syntax guidance drawer.

- **Top Line:** Ready-to-paste command with target IP pre-substituted. Press **[Enter]** or **[y]** to copy.
- **Second Line:** Tactical explanation and parameter tip.
- **Input Row:** Press **[:]** to focus. Type natural notes, commands (e.g. :w rockyou), or capture data.

TOP MACHINE STATUS STRIP — Instant context on current active target.

- Shows Target IP, Hostname, OS, Scope badge ([IN-SCOPE]).
- Live progress bar showing methodology completion percentage.
- **NEXT ACTION** recommendation highlighting the immediate next step.

3. Complete Keyboard Shortcuts Cheatsheet ("Muscle Memory Map")

CYBOX-S is 100% operational from the keyboard. Single-letter keys trigger immediate actions unless an input dialog is active.

Key	Category	Action & Operational Behavior
1, 2, 3, 4	Station	Switch immediately to Station 1 (Cockpit), 2 (Playbooks), 3 (Creds Matrix), or 4 (Loot).
w	Navigation	Cycle Panels: Sequentially rotates focus across all 5 panels in Cockpit without using Tab.
h / l	Navigation	Column Jump: Press h to jump to Sidebar (left); press l to jump to Workbench (right).
j / k	Navigation	Vim Navigation: Move highlight down (j) or up (k) in the active list or tree.
[/]	Targeting	Target Switcher: Switch immediately to previous [or next] target machine in your workspace.
b	Display	Sidebar Toggle: Collapse sidebar to give 100% horizontal width to Workbench (press b again to restore).
: (colon)	Console	Focus Command Bar: Instantly activates the bottom console input ready to type.
Esc	Console	Dismiss / Blur: Clears the command input and returns cursor focus directly to the workbench list.
Space	Triage / Reveal	• On Service: Cycle status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED]. • On Credential: Reveal/mask secret password. • In Station 3 Matrix: Cycle test state: [UNTESTED] → [VALID] → [PWN3D] → [INVALID].
, / .	Tool Carousel	Multi-Tool Recipe Carousel: When a service is highlighted, press . (next) or , (prev) to cycle alternative tool recipes in the console (e.g. feroxbuster → gobuster → nikto → curl).
Enter	Execute / Copy	• On Guidance / Service: Copies the previewed tool command to system clipboard. • In Station 3 Matrix: Compiles and copies ready-to-run spray command for highlighted credential & port.
y	Copy	Quick Copy: Copies the selected entity's primary value (IP address, port, password, or checklist command).
z	Layout	Zoom: Maximize the focused panel to full-screen; press z again to restore 4-quadrant layout.
T	Theme	Theme Picker: Open visual theme modal with 10 vibrant palettes (press d to set default).
/ or Ctrl+F	Search	Global Fuzzy Search: Search across targets, services, credentials, checklists, and notes.
r	Reference	Cheat Sheet: Open offline practical reference dialog for quick syntax inspection.
t	Fast Capture	Open Add Target dialog (or type :t <ip> [host] [os] in console).
s	Fast Capture	Open Add Service dialog (or type :s <port/proto> <service> in console).
c	Fast Capture	Open Add Credential dialog (or type :c <user:pass> [scope] in console).
n	Fast Capture	Open Add Note dialog (or type :n <text> in console).
f	Fast Capture	Open Add Finding / Vulnerability dialog (or type :f <title> in console).
m	Methodology	Open Methodology Picker dialog (or type :m <name> in console).
d	Manage	Delete highlighted entity (requires confirmation).

PRO-TIP: THE ZERO-MOUSE OPERATOR FLOW

1. Highlight any port in **SERVICES & PORTS** with j / k.
2. Watch the bottom console display the exact syntax with the target IP already substituted.
3. Press . to cycle alternative tools (e.g., feroxbuster → gobuster → nikto → curl).
4. Press Enter to copy the command directly to your terminal clipboard, switch to your tmux/terminal pane, and run it!
5. Press Space to mark the port [CHECKED] or [DEAD-END].

4. Bottom Console Syntax & Wordlist Accelerators

Console Command	Description & Operational Effect	Example Syntax
<code>:w <alias></code>	Wordlist Accelerator: Instantly copies standard Kali / SecLists paths to clipboard.	<code>:w rockyou :w common :w medium</code> <code>:w raft-d :w raft-f :w users</code>
<code>:t <ip> [host] [os]</code>	Add target machine with optional hostname and OS tag.	<code>:t 10.10.10.25 dc01.corp.local</code> Windows
<code>:s <port> <svc></code>	Record newly discovered port & service on active target.	<code>:s 445/tcp smb :s 8080 http</code>
<code>:c <user:pass> [scope]</code>	Record discovered credential with optional service scope.	<code>:c admin:Secret123! SMB :c</code> root:toor SSH
<code>:m <template></code>	Instantiate standard checklist template (replace or append).	<code>:m ejpt :m web :m smb append</code>
<code>:uflag / :rflag <hash></code>	Record user flag or root/admin flag into Station 4 Loot Ledger.	<code>:uflag 7a9e... :rflag f04b...</code>
<code>:st <where> / :cl <clue></code>	Record stuck point (rabbit hole) and breakthrough clue.	<code>:st stuck on wp-login :cl found</code> backup.zip
<code>:theme <name></code>	Switch theme live (e.g. sugary, caramel, midnight, cyber).	<code>:theme sugary :theme midnight</code>

5. Step-by-Step Practical Exam Workflow ("The Battle Plan")

Follow this methodical 5-phase execution cycle to maintain relentless momentum and pass calmly:

PHASE 1: SCOPING & HOST DISCOVERY

- 1. Network Route Verification:** Run `ip a` & `ip route` to identify your assigned lab IP and gateway subnet.
- 2. Subnet Ping Sweep:** Run `fping -a -g <SUBNET>/24 2>/dev/null` or `arp-scan --localnet`.
- 3. Add Targets to Cockpit:** For each live host, type `:t <IP>` in the console. Cycle between targets with `[ ]` and `]`.
- 4. Apply eJPT Checklist:** Press `[m]` and choose **eJPT Practical Pentest**. The status strip immediately shows your progress bar!

PHASE 2: PORT ENUMERATION & SERVICE TRIAGE

- 1. Full Port Scan:** Run full TCP scan: `nmap -p- -sS -T4 <TARGET_IP>`.
- 2. Record Open Ports:** Add discovered services quickly: `:s 80/tcp http, :s 445 smb, :s 22 ssh`.
- 3. Use Multi-Tool Carousel:** Move highlight to a service. Press `.` and `,` to cycle alternative inspection recipes in the bottom console! For HTTP, cycle between whatweb, feroxbuster, gobuster, nikto, and curl -I. Press `[Enter]` to copy.
- 4. Service Triage with [Space]:** As you test services, press `[Space]` to cycle: UNTESTED → [CHECKED] (done) → [DEAD-END] (no attack vector) → [DEFERRED] (test later).

PHASE 3: WEB & LOW-HANGING FRUIT EXPLOITATION

- 1. Inspect Anonymous / Default Access:** Check anonymous FTP (`:s 21`), null SMB shares (`:s 445`), and SNMP strings (`:s 161`).
- 2. Directory Fuzzing:** Type `:w common` or `:w medium` to copy the SecLists directory path. Paste directly into feroxbuster or gobuster.
- 3. Cross-Filtering Advantage:** When you highlight a service in SERVICES & PORTS, CYBOX-S automatically auto-scrolls the **Methodology** checklist to that exact service's testing steps!

PHASE 4: CREDENTIAL SPRAYING & LATERAL MOVEMENT (STATION 3)

- 1. Record Every Credential:** Whenever you find credentials (in web config, source code, or database), press `[c]` or type `:c user:password scope`.
- 2. Open Station 3 Matrix:** Press `[3]` to switch to the 2D Credential Spray Matrix.
- 3. Instant Spray Generation:** Highlight any cell and press `[Enter]`! CYBOX-S automatically compiles and copies the exact command to your clipboard (e.g. `netexec smb 10.10.10.20 -u admin -p 'Password123' or evil-winrm -i 10.10.10.20 -u admin -p 'Password123'`).
- 4. Record Verification State:** Run the command in your terminal. Press `[Space]` on the cell to cycle: [UNTESTED] → [VALID] → [PWN3D] → [INVALID]. Never test the same dead cred twice!

PHASE 5: PRIVESC, PIVOTING & EVIDENCE CAPTURE (STATION 4)

- 1. Pivoting & Route Addition:** If a target has dual NICs, run `ip route add <NEW_SUBNET> via <PIVOT_IP>` or setup Chisel / autoroute.
- 2. Privilege Escalation:** Switch to Station 2 (press `[2]`) and search Linux SUID, `sudo -l`, Cron jobs, or Windows token / service permissions.
- 3. Capture Flags & Answers:** Immediately type `:uflag <hash>` or `:rflag <hash>` when found.
- 4. Avoid Exam Amnesia:** Press `[4]` to open Station 4 Loot. Review your proof ledger to directly answer exam questions without re-exploiting!

THE 20-MINUTE ROTATION RULE (EXAM GOLDEN RULE)

If you have been analyzing a single service for more than 20 minutes without a new lead: **STOP**. Press `[Space]` on the service to mark it [DEFERRED]. Type `:stuck <service description>` to log where you are stuck, press `]` to rotate to another machine or port, and harvest low-hanging fruit elsewhere. 80% of exam failures occur because students spend 3 hours trapped down an unexploitable rabbit hole!

6. Station 3 Deep Dive: 2D Credential Spray Matrix

In multi-machine exams (like eJPTv2 / eCPPT), credential reuse across SSH, SMB, WinRM, and Web portals is the #1 lateral movement technique. Station 3 prevents credential amnesia by organizing credentials against in-scope authenticating services:

Credential (Row)	10.10.10.20:22 (SSH)	10.10.10.20:445 (SMB)	10.10.10.25:5985 (WinRM)	10.10.10.30:3389 (RDP)
admin : P@ssword123	[VALID]	[PWN3D]	[UNTESTED]	[INVALID]
guest : <empty>	[INVALID]	[VALID]	[INVALID]	[UNTESTED]
dbuser : dbpass2024	[UNTESTED]	[INVALID]	[UNTESTED]	[UNTESTED]

Navigation: Use arrow keys to select any cell. Press **[Enter]** to instantly copy the command (e.g. netexec smb 10.10.10.20 -u admin -p 'P@ssword123'). Press **[Space]** to cycle status: [UNTESTED] → [VALID] → [PWN3D] → [INVALID].

7. Built-In Methodology Templates (Type :m)

Template Alias	Methodology Category	Key Checklist Steps & Focus
:m ejpt	eJPT Practical Pentest	Full 12-step structured workflow: Scope recon → Host discovery → Full TCP scan → Service detection → Low-hanging fruit → Web fuzzing → Auth inspection → Exploitation → Pivoting → PrivEsc.
:m web	OWASP Web Assessment	Technology profiling (whatweb) → Content discovery (feroxbuster/gobuster) → Input parameter fuzzing → Authentication bypass → SQL Injection → File inclusion (LFI/RFI) → Upload forms.
:m smb	SMB & Active Directory	Null session shares (smbclient/smbmap) → User & group enumeration (rpcclient) → Password policy check → Anonymous signing audit → Vulnerability scan (MS17-010, ZeroLogon).
:m pivoting	Network Pivoting & Routing	Dual-homed adapter discovery (ip a) → Internal route table inspect → Proxychains & Chisel server setup → Port forwarding (socat) → Internal segment host sweep.
:m privesc_linux	Linux Local PrivEsc	sudo -l commands → SUID/SGID binaries (GTF0Bins) → Capabilities (getcap) → World-writable scripts & crontabs → Internal listening ports → Password files (/etc/shadow permissions).
:m privesc_windows	Windows Local PrivEsc	whoami /priv (SeImpersonate) → Unquoted service paths → Modifiable service binaries → Stored credentials (cmdkey /list) → AlwaysInstallElevated → SAM/SYSTEM registry backup.

8. Exam-Day Readiness & CLI Reference

EXAM START PROCEDURES:

1. Open terminal on your host or VM and start: cyb0x-s

2. Pick your favorite visual theme: press **[T]**, select palette (e.g. *Sugary*, *Midnight*, *Slate*, or *Cyber*), press **[d]** to save as default.

3. Import initial Nmap scan: type cyb0x-s import /path/to/scan.xml or fast-add targets with :t <ip>.

4. **Database Location:** Stored in ~/.local/share/cyb0x-s/worksheets.db. Completely local, persistent, and never transmits data outside.